

Availability is defined as the probability that a component (or system) is performing its intended function at a given time, t , on the assumption that it is operated and maintained as per the prescribed conditions. This is referred to as point availability and denoted by $A(t)$.

It is to be observed that reliability is concerned with failure-free operation up to time t , whereas availability is concerned with the capability to operate at the point of time t .

If $A(t)$ is the point availability of a component (or system), then

$$A(t_2 - t_1) = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} A(t) dt$$

is called the interval availability or mission availability.

In particular, the interval availability over the interval $(0, T)$ is

$$A(T) = \frac{1}{T} \int_0^T A(t) dt$$

Now $\lim_{T \rightarrow \infty} A(T)$ is called the steady-state or asymptotic or long-run availability and denoted by A or $A(\infty)$.

Formula for availability

Point availability $A(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t}$

Where λ = failure rate (or) Hazard function
 μ = Repair rate function

Interval availability on $[0, T]$,

$$A(T) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{(\lambda + \mu)^2 \times T} \{1 - e^{-(\lambda + \mu)T}\}$$

Steady State Availability

$$A(\infty) = \frac{\lambda}{\lambda + \mu} = \frac{MTTF}{MTTF + MTTR}$$

Example:

Reliability testing has indicated that a voltage inverter has a 6 month reliability of 0.87 without repair facility. If repair facility is made available with an MTTR of 2.2 months, compute the availability over the 6-month period. (Assume constant failure and repair rates)

Soln:- Given $R(6) = 0.87$
 $MTTR = \frac{1}{\mu} = 2.2 \Rightarrow \mu = 0.4545/\text{month}$
Find Interval availability for (0,6)

From $R(6) = 0.87 \Rightarrow e^{-\lambda 6} = 0.87 \Rightarrow \lambda = 0.0232/\text{month}$

Now, Interval availability for (0,6) is

$$A(T) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{(\lambda + \mu)^2 T} \{1 - e^{-(\lambda + \mu)T}\}$$

Here $T = 6$

$$\therefore A(6) = \frac{0.4545}{0.4545 + 0.0232} + \frac{0.0232}{(0.4545 + 0.0232)^2 6} \{1 - e^{-(0.0232 + 0.4545)6}\}$$

$$= 0.967$$

Example:

A new computer has a constant failure rate of 0.02 per day (assuming continuous use) and a constant repair rate of 0.1 per day.

Compute the interval availability for the first 30 days and the steady-state availability.

Soln:-

$$\lambda = 0.02; \mu = 0.1 \text{ \& } T = 30$$

Availability for the interval $(0, 30)$ is

$$\begin{aligned} A(30) &= \frac{\mu}{(\lambda + \mu)} + \frac{\lambda}{(\lambda + \mu)^2 \times T} \left\{ 1 - e^{-(\lambda + \mu) 30} \right\} \\ &= \frac{0.1}{0.12} + \frac{0.02}{(0.12)^2 \times 30} \left\{ 1 - e^{-0.12 \times 30} \right\} \\ &= 0.8784 \end{aligned}$$

$$A(\infty) = \frac{\frac{1}{\lambda}}{\frac{1}{\lambda} + \frac{1}{\mu}} = \frac{\frac{1}{0.02}}{\frac{1}{0.02} + \frac{1}{0.1}} = 0.8333$$